

Anders E. Nielsen

ENGINEERING LEADERSHIP · PLATFORM ENGINEERING · ML/AI · BAYESIAN STATISTICS

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I am an engineering leader who builds and scales the ML/AI and data platforms behind quantitative research, combining hands-on ownership of production infrastructure with a strong grounding in ML. I currently head Software and ML Engineering at Gubra, where my teams apply machine learning, and modern software practices to automate and scale scientific processes.

Experience

Head of Software & ML Engineering (Senior Engineering Manager)

Gubra

DIGITAL, DATA, AND IT

June 2026 to Present

- Department manager for two teams, 14 people total, owning the software and ML engineering organization behind Gubra's drug discovery and scientific automation platforms.
- Organized the department into two functionally distinct teams: one ML/AI-focused, one software-development-focused. Set up complementary delivery processes, Scrum and Kanban, across them, each collaborating with dedicated Product Owners.
- Designed and launched an organization-wide system for deciding, documenting, and enforcing engineering policies, covering CI/CD, technology choices, incident response, code conventions, and more.
- Leading a shift from home-grown software to commercial off-the-shelf solutions for undifferentiated capabilities, reserving developer capacity for integrations and business differentiating work.
- Drafting Gubra's AI and data strategy, shifting the organization toward a dedicated platform engineering strategy and team that other engineers and scientists build on.
- Navigating organizational change as the company scales.

Engineering Manager

Gubra

SCIENTIFIC AUTOMATION

May 2025 to May 2026

- Built the Scientific Automation team from scratch, establishing the technical practices and delivery cadences that still underpin both teams I lead today.
- Grew the team from 3 to 7 engineers.
- Applied ML, AI, and good software engineering practices to automate and scale scientific processes including drug discovery, digital histopathology, and mass spectrometry.
- Manage risk in production ML by keeping humans in the loop: for our scientific image-analysis pipelines, human reviewers pass/fail the most uncertain predictions and can trigger retraining.

Senior Developer

Gubra

SOFTWARE SOLUTIONS IN COMPUTATIONAL BIOLOGY

May 2021 to April 2025

- Responsible for the data and app infrastructure for Gubra's drug discovery platform, working across the full stack: Postgres for data storage, FastAPI for backend, R Shiny for frontend, GitHub Actions for CI/CD, and Docker Swarm for deployment.
- Designed and implemented a data-based design-and-analysis loop for drug development using Bayesian statistics and experimental design.
- Applied Large Language Models (LLMs) to peptide-based drug design, an early production use of generative AI on the platform.
- Applied ML, AI, and good software engineering practices to automate and scale scientific processes including drug discovery, digital histopathology, and mass spectrometry.
- Built pipelines for automatic data analysis, ensuring uniform, comparable treatment of results across hundreds of experiments.
- Introduced Scrum to the team and served as Scrum Master for years, establishing a sprint cadence and facilitating continuous improvement retrospectives.

Data Scientist

Novo Nordisk

MANUFACTURING INTELLIGENCE AND SUPPLY CHAIN

September 2017 to April 2021

- Involved in hiring 15+ data professionals, personally leading 100+ interviews.
- Increased overall equipment efficiency of an assembly line by combining Design of Experiments with Gaussian Process Regression and Bayesian optimisation. Skills applied: ML, A/B testing, Python, Bayesian statistics.
- Predictive maintenance on brownfield manufacturing equipment. Skills used: AWS Greengrass, AWS IoT, CI/CD with Azure DevOps.
- Created dashboards for factory planning. Skills applied: Docker, Tableau, Streamlit, visualisation, Python.
- Built a regulatory framework for applying machine learning to quality-critical pharma processes. Skills applied: GxP, models in production, model validation, model monitoring.

Intern

Embassy of Denmark in South Korea

INNOVATION CENTRE DENMARK

July 2016 to January 2017

- Collected market intelligence, facilitated Korean-Danish technology collaboration, arranged and executed large events, and had various diplomatic tasks.

Education

Technical University of Denmark (DTU)

Lyngby, Denmark

M.SC. BIOINFORMATICS AND SYSTEMS BIOLOGY ENGINEERING

2015-08 to 2017-08

- Thesis on applying Deep Learning techniques to identify potential enzyme sequences in large metagenomic datasets.

Technical University of Denmark (DTU)

Lyngby, Denmark

B.SC. BIOTECHNOLOGY ENGINEERING

2012-08 to 2015-07

Skills

Engineering Leadership	Team Leadership, Team Building, Org Design, Hiring, Agile/Scrum, Engineering Governance, Change Management, Risk-Based Decision Making, Platform & AI Strategy
Bayesian Statistics	Bayesian Statistics, Experimental Design, Gaussian Processes, Multivariate Statistics
Platform Engineering	Docker, AWS, CI/CD, Postgres, FastAPI, MSSQL, DynamoDB
Machine Learning / AI	LLMs, Deep Learning, SciKit, PyTorch
Programming & Scripting	Python, R, Shiny, Unix
Computational Biology	Metagenomics, Peptide Design, NGS
Languages	Fluent in Danish, English, and Korean